

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			817	USE ONLY FOR TIE IN
20										826	TO BE USED FOR GATHERING PROJECTS ONLY
20							***** MODULAR VALVES *****			B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
30	U04		MODULAR DBB	K15AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B4	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AC02
40	U04		MODULAR DBB	K15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
50	U04	VAV	MODULAR DBB	K09AC06	3	3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO FULL SIZE TYPE 2BV	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
60	U04		MODULAR DBB	M15AC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
70	U04		MODULAR DBB	M15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
80	U04	VAW	MODULAR DBB	M15EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
90	U04	VAW	MODULAR DBB	M09EC02	1	1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
100		VB2					***** BALL VALVES *****			VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
100										VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
110	U04		BALL	A15AC05	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB LO	0	0		
120	U04		BALL	A15AC06	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL RB LO	0	0		
130	U04		BALL	A09AC05	3	3	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB LO	0	0		
140	U04		BALL	A09AC06	3	3	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB LO	0	0		
150	U04		BALL	A09AC07	4	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0		
160	U04	826	BALL	A09AC08	16	16	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
170	U04		BALL	A09AC08	4	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
180	U04	817	BALL	A09CC07-K5B	12	12	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0		
190	U04	817	BALL	A09CC08-K5B	16	16	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
200	U04	817	BALL	A09CC08-K5B	12	12	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
210	U04		BALL	B15AC01	0.75	1.5	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING SOFT FB LO	0	0		
220	U04	B4	BALL	B15AC02	0.75	1.5	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING METAL FB LO	0	0		
230	U04	VA5	BALL	B15AC05	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB LO	0	0		
240	U04	VA5	BALL	B15AC06	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL FB LO	0	0		
250	U04	VA5	BALL	B09AC07	3	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
260	U04	VA5	BALL	B09AC08	3	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
270	U04	817	BALL	B09CC07-K5B	12	12	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
280	U04	817	BALL	B09CC08-K5B	12	12	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
290							***** RISING STEM BALL VALVES *****				
290											
300											
300	U04	VA5	BALL	R15AC06	2	2	BALL RISING STEM 1500# RF LTCS A352 LCC INCONEL TRIM METAL RB HO	0	0		
310	U04	VA5	BALL	R09AC08	6	6	BALL RISING STEM 900# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
320	U04	VA5	BALL	R09AC08	8	8	BALL RISING STEM 900# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
330	U04	VA5	BALL	R09AC06	4	4	BALL RISING STEM 900# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0		
340							***** THROUGH CONDUIT GATE VALVES *****				
340											
350											
350			GATE	W09AC70	3	4	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0		
360			GATE	W09AC71	6	10	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
370	U04	817	GATE	W09CC71-K5B	12	12	GATE 900# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0		
380							***** GLOBE VALVES *****				
380											
390											
390	U04		GLOBE	L15AC48	0.75	1.5	GLOBE 1500# RF LTCS A350 LF2 CL.1 INCONEL OS&Y HW	0	0		
400	U04		GLOBE	L15AC41	2	2	GLOBE 1500# RF LTCS A352 LCC INCONEL OS&Y HW	0	0		
410			GLOBE	L09AC41	3	3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
420			GLOBE	L09AC42	4	8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
430							***** CHECK VALVES *****				
430											
440											
440	U04	VC1	CHECK	C15AC50	0.75	1.5	CHECK PISTON 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM	0	0		
450	U04		CHECK	C15AC45	2	2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC INCONEL TRIM THROU BOLTED	0	0		
460			CHECK	C09AC45	3	9	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
470	U04	826	CHECK	C09AC52	16	16	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16	0	0		
480			CHECK	C09AC52	10	10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16	0	0		
490	U04	817	CHECK	C09CC52-K5B	12	12	CHECK DUAL DBL FLG 900# CF LTCS A352 LCC TRIM 16	0	0		
500	U04	817	CHECK	C09CC52-K5B	16	16	CHECK DUAL DBL FLG 900# CF LTCS A352 LCC TRIM 16	0	0		
510							***** PIPE ABOVE GROUND *****				
510											
520											

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 1 of 4	900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
520			PIPE	5358512	0.75	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0	
530			PIPE	5358515	2	3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0	
540										
540			PIPE	9005240	4	4	LINEPIPE SMLS WT= 8.56 MM API-5L X60 BE	0	0	
550			PIPE	9001493	6	6	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE	0	0	
560	U04		PIPE	9001495	8	8	LINEPIPE SMLS WT= 12.70 MM API-5L X60 BE	0	0	
570			PIPE	9001491	10	10	LINEPIPE SMLS WT = 14.27 MM API-5L X60 BE	0	0	
580			PIPE	9005244	12	12	LINEPIPE SMLS WT = 15.88 MM API-5L X60 BE	0	0	
590			PIPE	9005243	16	16	LINEPIPE SMLS WT= 19.05MM API-5L X60 BE	0	0	
600										
600							***** PIPE BELOW GROUND *****			
610										
610			PIPE	9001311CTD	6	6	LINEPIPE SMLS WT= 7.92 MM API-5L X60 BE COATED	0	0	
620										
620							*****FITTINGS ABOVE & BELOW GROUND*****			
630										
630			ELBOW	5392822	0.75	3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0	
640			ELBOW	9006567	16	16	ELL BW 45 DEG LR WT 19.05MM A860 WPHY 60	0	0	
650	U04		ELBOW	9006590	8	8	ELL BW 45 DEG LR WT 12.70MM A860 WPHY 60	0	0	
660	U04		ELBOW	9006581	12	12	ELL BW 45 DEG LR WT 15.88MM A860 WPHY 60	0	0	
670	U04		ELBOW	9009597	4	4	ELL BW 45 DEG LR WT 8.56 MM A860 WPHY 60	0	0	
680	U04		ELBOW	9006583	6	6	ELL BW 45 DEG LR WT 9.53MM A860 WPHY 60	0	0	
690			ELBOW	5392816	0.75	3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0	
700			ELBOW	9006594	4	4	ELL BW 90 DEG LR WT 8.56 MM A860 WPHY 60	0	0	
710			ELBOW	9006469	12	12	ELL BW 90 DEG LR WT 15.88 MM A860 WPHY 60	0	0	
720			ELBOW	9006568	16	16	ELL BW 90 DEG LR WT 19.05 MM A860 WPHY 60	0	0	
730	U04		ELBOW	9006479	8	8	ELL BW 90 DEG LR WT 12.70 MM A860 WPHY 60	0	0	
740	U04		ELBOW	9006480	10	10	ELL BW 90 DEG LR WT 14.27 MM A860 WPHY 60	0	0	
750	U04		ELBOW	9006471	6	6	ELL BW 90 DEG LR WT 9.53 MM A860 WPHY 60	0	0	
760										
760			RED CONC	5392827	0.75	2	REDUCER CONC SCH 160 LTCS A420 WPL6	1	3	
770			RED CONC	9006924	1.5	3	REDUCER CONC WT 8.56MM / SCH 160 A860 WPHY 60	4	4	
780			RED CONC	9006925	3	3	REDUCER CONC WT 9.53MM / SCH 160 A860 WPHY 60	6	6	
790			RED CONC	9006900	8	8	REDUCER CONC WT 14.27MM / 12.7MM LTCS A860 WPHY 60	10	10	
800	U04		RED CONC	9002214	6	6	REDUCER CONC WT 12.7MM / 9.53MM A860 WPHY 60	8	8	
810	U04			9006937	12	12	REDUCER CONC WT 19.05MM / 15.88MM LTCS A860 WPHY-60	16	16	
820			RED ECC	5392828	0.75	2	REDUCER ECC SCH 160 LTCS A420 WPL6	1	3	
830			RED ECC	9004273	1.5	3	REDUCER ECC WT 8.56MM / SCH 160 LTCS A860 WPHY-60	4	4	
840			RED ECC	9004274	3	3	REDUCER ECC WT 9.53 MM / SCH 160 LTCS A860 WPHY-60	6	6	
850			RED ECC	9002410	8	8	REDUCER ECC WT 14.27 MM / 12.70 MM LTCS A860 WPHY-60	10	10	
860	U04		RED ECC	9004277	4	4	REDUCER ECC WT 12.70 MM / 8.56 MM LTCS A860 WPHY-60	8	8	
870	U04		RED ECC	9002405	6	6	REDUCER ECC WT 12.70 MM / 9.53 MM LTCS A860 WPHY-60	8	8	
880	U04		RED ECC	9002106	8	8	REDUCER ECC WT 19.05MM / 12.70MM LTCS A860 WPHY-60	16	16	
890	U04		RED ECC	9002048	10	10	REDUCER ECC WT 19.05MM / 14.27MM LTCS A860 WPHY-60-W	16	16	
900	U04		RED ECC	9004285	12	12	REDUCER ECC WT 19.05MM / 15.88MM LTCS A860 WPHY-60	16	16	
910	U04		RED ECC	9004275	4	4	REDUCER ECC WT 9.53 MM / 8.56 MM LTCS A860 WPHY-60	6	6	
920										
920			TEE	5392825	0.75	3	TEE BW SCH 160 LTCS A420 WPL6	0	0	
930			TEE	9004200	12	12	TEE BW WT 15.88MM A860 WPHY 60	0	0	
940	U04		TEE	9006633	8	8	TEE BW WT 12.7 MM A860 WPHY 60	0	0	
950	U04		TEE	9006486	10	10	TEE BW WT 14.27MM A860 WPHY 60	0	0	
960	U04		TEE	9004212	16	16	TEE BW WT 19.05MM A860 WPHY 60-W	0	0	
970	U04		TEE	9004210	6	6	TEE BW WT 9.53MM A860 WPHY 60-W	0	0	
980			TEE RED	5392821	0.75	2	TEE RED BW SCH 160 LTCS A420 WPL6	1	3	
990			TEE RED	9006662	2	3	TEE RED BW WT 8.56 MM/SCH 160 LTCS A860 WPHY 60	4	4	
1000			TEE RED	9006665	3	3	TEE RED BW WT 9.53 MM/SCH 160 MM LTCS A860 WPHY 60	6	6	
1010			TEE RED	9004207	6	6	TEE RED BW WT 15.88 MM/9.53 MM LTCS A860 WPHY 60	12	12	
1020			TEE RED	9006634	6	6	TEE RED BW WT 19.05 MM/9.53 MM LTCS A860 WPHY 60	16	16	
1030	U04		TEE RED	9006643	6	6	TEE RED BW WT 12.70 MM/9.53 MM LTCS A860 WPHY 60	8	8	
1040	U04		TEE RED	9006622	8	8	TEE RED BW WT 14.27 MM/12.7 MM LTCS A860 WPHY 60	10	10	
1050										
1050			CAP	5392826	0.75	3	CAP BW SCH 160 LTCS A420 WPL6	0	0	
1060			CAP	9005200	4	4	CAP BW WT 8.56 MM LTCS A860 WPHY-60	0	0	
1070			CAP	9005060	6	6	CAP BW WT 9.53 MM LTCS A860 WPHY-60	0	0	
1080			CAP	9005067	8	8	CAP BW WT 12.70 MM LTCS A860 WPHY-60	0	0	
1090			CAP	9005068	10	10	CAP BW WT 14.27 MM LTCS A860 WPHY-60	0	0	
1100			CAP	9005069	12	12	CAP BW WT 15.88 MM LTCS A860 WPHY-60	0	0	
1110			CAP	9005072	16	16	CAP BW WT 19.05 MM LTCS A860 WPHY-60	0	0	
1120										
1120							***** INDUCTION BENDS BELOW GROUND*****			
1130										
1130			INDUC BEND	9001815CTD	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0	

	TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
1140			INDUC BEND	9001825CTD	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1150			INDUC BEND	9001835CTD	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1160	U04		INDUC BEND	9001842CTD	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 10.97 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1170	U04		INDUC BEND	9001841CTD	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 10.97 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1180	U04		INDUC BEND	9007003CTD	6	6	INDUCTION BEND 45DEG X 10D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1190	U04		INDUC BEND	9007018CTD	6	6	INDUCTION BEND 45DEG X 15D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1200	U04		INDUC BEND	9007033CTD	6	6	INDUCTION BEND 90 DEG X 10D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1210	U04		INDUC BEND	9007045CTD	6	6	INDUCTION BEND 90 DEG X 15D RADIUS WT 7.92 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1220	U04		INDUC BEND	9001840CTD	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 10.97 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0					
1230							***** PIPE BELOW GROUND *****							
1230														
1240														
1240	U04		PIPE	9001487CTD	6	6	LINEPIPE SMLS WT = 8.74 MM API-5L X60 BE COATED	0	0					
1250	U04		PIPE	9001493CTD	6	6	LINEPIPE SMLS WT = 9.53 MM API-5L X60 BE COATED	0	0					
1260	U04		PIPE	9001502CTD	8	8	LINEPIPE SMLS WT= 10.97 MM API-5L X60 BE COATED	0	0					
1270														
1270							*****INDUCTION BENDS ABOVE GROUND*****							
1280														
1280	U04		INDUC BEND	9005333	8	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0					
1290	U04		INDUC BEND	9001837	6	6	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0					
1300	U04		INDUC BEND	9005327	8	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0					
1310	U04		INDUC BEND	9001827	6	6	INDUCTION BEND 45 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0					
1320	U04		INDUC BEND	9005321	8	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 12.70 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0					
1330	U04		INDUC BEND	9001817	6	6	INDUCTION BEND 90 DEG X 5D RADIUS WT 9.53 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0					
1340														
1340							***** OLETS ABOVE GROUND *****							
1350														
1350			OLETS	9006819	0.75	1.5	WELDOLET WT 8.56 MM / SCH 160 LTCS A694 F60	4	4					
1360			OLETS	9002457	0.75	2	WELDOLET WT 9.53 MM/SCH 160 WT LTCS A694 F60	6	6					
1370			OLETS	9002466	0.75	3	WELDOLET WT 12.70 MM/ SCH 160 WT LTCS A694 F60	8	8					
1380			OLETS	9002470	0.75	3	WELDOLET WT 14.27 MM/ SCH 160 WT LTCS A694 F60	10	10					
1390			OLETS	9002085	0.75	3	WELDOLET WT 15.88 MM/ SCH 160 LTCS A694 F60	12	12					
1400			OLETS	9004335	4	4	WELDOLET WT 15.88 MM/ 8.56 MM LTCS A694 F60	12	12					
1410			OLETS	9004336	0.75	3	WELDOLET WT 17.48 MM/SCH 160 LTCS A694 F60	14	14					
1420			OLETS	9004337	4	4	WELDOLET WT 17.48 MM/8.56 MM LTCS A694 F60	14	14					
1430			OLETS	9002089	0.75	3	WELDOLET WT 19.05 MM/ SCH 160 LTCS A694 F60	16	16					
1440			OLETS	9004338	4	4	WELDOLET WT 19.05 MM/ 8.56 MM LTCS A694 F60	16	16					
1450			OLETS	9004339	6	6	WELDOLET WT 19.05 MM/ 9.53 MM LTCS A694 F60	16	16					
1460														
1460	U04		OLETS	5504170	1	1	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2 CL.1	2	2					
1470			OLETS	9002838	0.75	2	NIPOFLANGE 1500# RF WT 12.70 MM/ SCH 160 LTCS A694 F60	8	8					
1480			OLETS	9002839	0.75	2	NIPOFLANGE 1500# RF WT 14.27 MM/ SCH 160 LTCS A694 F60	10	10					
1490			OLETS	9004322	0.75	2	NIPOFLANGE 1500# RF WT 15.88 MM/ SCH 160 LTCS A694 F60	12	12					
1500			OLETS	9002842	0.75	2	NIPOFLANGE 1500# RF WT 19.05 MM/ SCH 160 LTCS A694 F60	16	16					
1510	U04		OLETS	9004319	0.75	2	NIPOFLANGE 1500# RF WT 8.56 MM/ SCH 160 LTCS A694 F60	4	4					
1520	U04		OLETS	9002875	0.75	2	NIPOFLANGE 1500# RF WT 9.53 MM/ SCH 160 LTCS A694 F60	6	6					
1530														
1530							***** FLANGES ABOVE GROUND*****							
1540														
1540	U04		FLANGE	5557397	0.75	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0					
1550	U04		FLANGE	5557145	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 160	0	0					
1560			FLANGE	9004327	4	4	FLG WN 900# RF LTCS A694 F60 WT 8.56 MM	0	0					
1570			FLANGE	9004328	6	6	FLG WN 900# RF LTCS A694 F60 WT 9.53 MM	0	0					
1580			FLANGE	9002849	8	8	FLG WN 900# RF LTCS A694 F60 WT 12.70MM	0	0					
1590			FLANGE	9002850	10	10	FLG WN 900# RF LTCS A694 F60 WT 14.27MM	0	0					
1600	U04	817	FLANGE	9002858	12	12	FLG WN 900# RF LTCS A694 F60 WT 15.88MM	0	0					
1610	U04	826	FLANGE	9002853	16	16	FLG WN 900# RF LTCS A694 F60 WT 19.05MM	0	0					
1620	U04	817	FLANGE	9006758	4	4	FLG WN 900# RTJ LTCS A694 F60 WT 8.56MM	0	0					
1630	U04	817	FLANGE	9006726	6	6	FLG WN 900# RTJ LTCS A694 F60 WT 9.53MM	0	0					
1640	U04	817	FLANGE	9006150	12	12	FLG COMPACT 900# RJ A694 F60 WT 15.88 MM	0	0					
1650	U04	817	FLANGE	9006154	16	16	FLG COMPACT 900# RJ A694 F60 WT 19.05 MM	0	0					
1660	U04		BLIND	5615616	0.75	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0					
1670	U04		BLIND	5615616	0.75	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0					
1680	U04		BLIND	5615605	3	3	FLG BLIND 900# RF LTCS A350 LF2 CL.1	0	0					
1690	U04		BLIND	9002863	4	16	FLG BLIND 900# RF LTCS A694 F60	0	0					
1700	U04	817		5615606	8	8	FLG BLIND 900# RTJ LTCS A350 LF2 CL.1	0	0					

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	900K5B

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1710	U04	817	BLIND	9619775	12	16	FLG COMPACT BLIND 900# RJ LTCS A694 F60 LINE CLASS 900K5B	0	0	
1720	U04	FA8	TEE RED	5599704	0.5	1	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2 CL.1	0.75	1.5	
1730	U04	817	FLANGE	5557511	2	2	FLG WN 1500# RTJ LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
1740	U04	817	FLANGE	5557281	3	3	FLG WN 900# RTJ LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
1750	U04	817	FLANGE	9006719	8	8	FLG WN 900# RTJ LTCS A694 F60 WT 12.70MM	0	0	
1760	U04	817	FLANGE	9006716	10	10	FLG WN 900# RTJ LTCS A694 F60 WT 14.27MM	0	0	
1770	U04	817	FLANGE	9006718	12	12	FLG WN 900# RTJ LTCS A694 F60 WT 15.88MM	0	0	
1780										
1780			FLANGE	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
1790			FLANGE	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
1800										
1800							***** GASKETS *****			
1810										
1810			GASKET	5672251	0.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1820	U04		GASKET	5673117	3	16	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1830	U04	817	GASKET	5673445	12	16	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 900K5B	0	0	
1840	U04	817	GASKET	5674491	3	12	GASKET 900# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
1850	U04	817	GASKET	5674492	2	2	GASKET 1500# RTJ SOFT IRON TEFLON COATED OCTAGONAL RING	0	0	
1860										
1860							***** BOLTING *****			
1870										
1870	U04		BOLTING	5675770	0.5	16	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1880										
1880		SB6	BOLTING	5675675	12	16	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1890										
1890							***** BLINDS & SPACERS *****			
1900										
1900			SPEC BLIND	5749595	0.75	2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70	0	0	
1910			SPEC BLIND	5749596	3	8	SPECTACLE BLIND 900# RF LTCS A516 GR.70	0	0	
1920	U04	817	SPEC BLIND	5749591	4	4	SPECTACLE BLIND 900# RTJ FEMALE LTCS A516-GR70	0	0	
1930										
1930			PADDLE	5750102	10	10	PADDLE BLANK 900# RF LTCS A516 GR.70	0	0	
1940			PADDLE	5750202	10	10	PADDLE SPACER 900# RF LTCS A516 GR.70	0	0	
1950	U04	817	PADDLE	5750222	12	16	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70	0	0	
1960	U04	817	PADDLE	5750228	12	16	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70	0	0	
1970										

	TENGINZCHEVROIL TCO TENGINZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - SOUR SERVICE ALL WELDS ABOVE GROUND SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 MAX. DESIGN PARAMETER = 132 BARG. @ 90 DEG.C. (LIMITED BY PIPE WT) FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 90 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	900K5B